Calculus with Parametric and Polar Curves

ANSWER KEY

Section 1: Derivatives of Parametric Curves

1 Given $x(t) = t^3 - 3t$ and $y(t) = t^2 + 1$, find $\frac{dy}{dx}$ in terms of t.

a. $\frac{2t}{3t^2 - 3}$ ✓

b. $\frac{3t^2 - 3}{2t}$

c. $\frac{2t}{3t^2}$

d. $\frac{2}{3t}$

$\frac{dy}{dx} = \frac{y'(t)}{x'(t)} = \frac{2t}{3t^2 - 3}$.

2 Find the equation of the tangent line to the parametric curve $x = 2\cos t$, $y = 3\sin t$ at $t = \pi/4$.

a. $y - \frac{3\sqrt{2}}{2} = -\frac{3}{2}(x - \sqrt{2})$ ✓

b. $y - \sqrt{2} = -\frac{2}{3}(x - \frac{3\sqrt{2}}{2})$

c. $y = -\frac{3}{2}x$

d. $y - 3 = -(x - 2)$

At $t = \pi/4$: $x = \sqrt{2}$, $y = 3\sqrt{2}/2$. $dx/dt = -2\sin t = -\sqrt{2}$; $dy/dt = 3\cos t = 3\sqrt{2}/2$. Slope $\frac{dy}{dx} = \frac{3\sqrt{2}/2}{-\sqrt{2}} = -\frac{3}{2}$. Line: $y - \frac{3\sqrt{2}}{2} = -\frac{3}{2}(x - \sqrt{2})$.

3 State the formula for the second derivative $\frac{d^2y}{dx^2}$ of a parametric curve.

a. $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}}$ ✓

b. $\frac{d^2y}{dx^2} = \frac{y''(t)}{x''(t)}$

c. $\frac{d^2y}{dx^2} = \frac{y''(t)}{[x'(t)]^2}$

d. $\frac{d^2y}{dx^2} = \frac{d}{dt}\left[\frac{dy}{dx}\right]$

By the chain rule: $\frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{dy}{dx}\right] = \frac{\frac{d}{dt}[dy/dx]}{dx/dt}$.

Section 2: Arc Length of Parametric Curves

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4 State the formula for the arc length of a parametric curve from $t = a$ to $t = b$.

a. $L = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt$ ✓

b. $L = \int_a^b (x'(t) + y'(t))\, dt$

c. $L = \int_a^b \sqrt{1 + [y'(t)]^2}\, dt$

d. $L = \int_a^b \sqrt{[x(t)]^2 + [y(t)]^2}\, dt$

$ds = \sqrt{dx^2 + dy^2} = \sqrt{(x'(t))^2 + (y'(t))^2}\, dt$, so $L = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt$.

5 Find the arc length of $x = 3\cos t, y = 3\sin t$ for $0 \le t \le 2\pi$.

a. 6π ✓

b. 3π

c. 9π

d. 12π

$x' = -3\sin t, y' = 3\cos t \Rightarrow \sqrt{9\sin^2 t + 9\cos^2 t} = 3$. $L = \int_0^{2\pi} 3\, dt = 6\pi$.

6 Find the arc length of $x = t^2, y = \frac{2}{3}t^3$ from $t = 0$ to $t = 1$.

a. $\frac{2}{3}(2\sqrt{2} - 1)$ ✓

b. $\frac{1}{3}(2\sqrt{2} - 1)$

c. $\sqrt{2} - 1$

d. $2\sqrt{2}$

$x' = 2t$, $y' = 2t^2 \Rightarrow \sqrt{4t^2 + 4t^4} = 2t\sqrt{1 + t^2}$. Let $u = 1 + t^2, du = 2t\, dt$.
$\int_1^2 u^{1/2}\, du = [\frac{2}{3}u^{3/2}]_1^2 = \frac{2}{3}(2\sqrt{2} - 1)$.

Section 3: Slopes and Tangents of Polar Curves

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7 Given $r = f(\theta)$, how are the Cartesian coordinates related to polar coordinates?

a. $x = r\cos\theta, y = r\sin\theta$ ✓

b. $x = r\sin\theta, y = r\cos\theta$

c. $r = x + y, \theta = y/x$

d. $x = r^2\cos\theta, y = r^2\sin\theta$

The standard polar to Cartesian conversion is $x = r\cos\theta = f(\theta)\cos\theta$ and $y = r\sin\theta = f(\theta)\sin\theta$.

8 Find the slope $\frac{dy}{dx}$ of the polar curve $r = 2$ at $\theta = \pi/4$.

a. -1 ✓

b. 1

c. 0

d. Undefined

Since $r = 2$ is a circle centered at the origin, the radius vector at $\theta = \pi/4$ has slope $\tan(\pi/4) = 1$. The tangent line is perpendicular to the radial line, so its slope is -1.

9 At what angles $\theta \in [0, \pi)$ does the cardioid $r = 1 + \cos\theta$ pass through the pole ($r = 0$)?

a. $\theta = \pi$ ✓

b. $\theta = \pi/2$

c. $\theta = 0$

d. $\theta = 3\pi/4$

$r = 0 \Rightarrow 1 + \cos\theta = 0 \Rightarrow \cos\theta = -1 \Rightarrow \theta = \pi$.

Section 4: Area in Polar Coordinates

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10 State the formula for the area enclosed by a polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$.

a. $A = \frac{1}{2}\int_{\alpha}^{\beta} [f(\theta)]^2\, d\theta$ ✓

b. $A = \int_{\alpha}^{\beta} f(\theta)\, d\theta$

c. $A = \pi\int_{\alpha}^{\beta} [f(\theta)]^2\, d\theta$

d. $A = \frac{1}{2}\int_{\alpha}^{\beta} f(\theta)\, d\theta$

An infinitesimal sector with central angle $d\theta$ has area $dA = \frac{1}{2}r^2 d\theta$, giving $A = \frac{1}{2}\int_{\alpha}^{\beta} r^2\, d\theta$.

11 Find the area enclosed by one petal of the four-petaled rose $r = 2\cos(2\theta)$.

a. $\frac{\pi}{2}$ ✓

b. π

c. 2π

d. $\frac{\pi}{4}$

One petal spans from $-\pi/4$ to $\pi/4$.
$A = \frac{1}{2}\int_{-\pi/4}^{\pi/4} 4\cos^2(2\theta)\, d\theta = 2\int_{-\pi/4}^{\pi/4} \frac{1+\cos(4\theta)}{2}\, d\theta = \int_{-\pi/4}^{\pi/4} (1+\cos 4\theta)\, d\theta = [\theta + \frac{\sin 4\theta}{4}]_{-\pi/4}^{\pi/4} = \frac{\pi}{4} - (-\frac{\pi}{4}) = \frac{\pi}{2}$.

12 Find the area of the region inside the circle $r = 4\sin\theta$.

a. 4π ✓

b. 16π

c. 2π

d. 8π

Tracing the circle requires $\theta \in [0, \pi]$. $A = \frac{1}{2}\int_{0}^{\pi} 16\sin^2\theta\, d\theta = 8\int_{0}^{\pi} \frac{1-\cos 2\theta}{2}\, d\theta = 4[\theta - \frac{\sin 2\theta}{2}]_{0}^{\pi} = 4\pi$.

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